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CleanEnviro Solutions

A PUBLICATION OF THE CleanEnviro Summit Singapore



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CLEANENVIRO SUMMIT SINGAPORE

With waste volumes of all kinds increasing in the Asia Pacific region, Singapore believes that it is timely for a serious dialogue to begin on how the region's cities will overcome the challenges of waste management. While much of the current focus is on water and energy issues, waste is an important part of the equation. We cannot solve one part of this equation without solving the other parts. The water-waste-energy nexus will therefore be a key challenge for Asia's growing urban regions.

The inaugural CleanEnviro Summit Singapore was the global platform for leaders, senior government officials and policymakers, regulators and industry captains to identify, develop and share practical solutions to address environmental challenges for tomorrow's cities. CleanEnviro Summit Singapore 2012 facilitated the sharing of insights on the latest environmental market trends through its plenary sessions and business forums. It was comprised of the the following three pillar events:

- **Clean Environment Leaders Summit**
  In Conversation with Ms Helen Clark, United Nations Development Programme Administrator.
  Joint Opening Plenary on "Sustainable Development – Our Cities, Water and the Environment".
  Clean Environment Leaders Plenary Sessions, organised in partnership with the United Nations Environment Programme.
- **Clean Environment Regulator's Roundtable**
- **WasteMET Asia**, the exhibition and conference segment showcased the latest innovations in waste management, environmental technology and recycling solutions for Asia.

Due to the outstanding success of the 2012 inaugural CleanEnviro Summit Singapore, the National Environment Agency (NEA) is already planning for the event's next showing on 1–4 June 2014. For more information, please visit www.cleanenvirosummit.sg for programme announcements and conference content.

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TOWARDS GREATER RESOURCE EFFICIENCY IN ASIA



Cities have become the focal point to solve today's solid waste challenges

What can be done to make Asia more resource efficient and hence more sustainable? Representatives from Asian governments and the World Bank discuss solutions on how best to achieve this.

Governments across Asia face intense pressure to curb their use of natural resources and reduce levels of solid waste and air pollution. Besides the obvious "use less and recycle more", it is widely acknowledged that in order for this goal to be achieved, leaders from across the region must focus on two areas — governance and tangible solutions.

GOVERNANCE

For many, it starts with sound policymaking and regulation. An advocate of this view is Ms Pamela Cox, Vice President of East Asia and the Pacific at the World Bank, who believes there are three important decisions that must be made in order to tackle today's resource efficiency challenges. "First, countries and cities need to decide for themselves what makes sense and what is the best solution for their particular circumstances, rather than use models that have been used by others," claims Ms Cox.

"Secondly, they need to make rational, efficient and sustainable decisions that involve both the private sector and their citizens, as without the buy-in of both they will never slow resource consumption. And thirdly, they need to find more innovative sources of financing, in order to support new initiatives. Rather than solely rely on governments or big corporations, [funds] must additionally come from the SME sector or even the general public," says Ms Cox.

In addition, Ms Cox notes, "In the World Bank's experience, decisions about how to best slow resource consumption and increase waste reuse are better made locally rather than nationally."

This is indeed how such decisions are made in Japan, as explained by His Excellency Mr Tatsushi Terada, Vice Minister for Global Environmental Affairs, Ministry of the Environment, Japan.

"In Japan, local governments plan and coordinate environmental projects. The national government's role is only to regulate and proactively share

these experiences between its different regions," said Mr Terada.

Mr Terada argues that planning locally enables greater buy-in from businesses and communities in sectors such as recycling and emissions reduction. He also claims that by planning locally, public acceptance of environmental solutions is currently at an all-time high.

However, some nations argue that a more centralised approach to resource efficiency works better. This is the stance taken by His Excellency Dr Yoon Jong Soo, Vice Minister for the Environment, Republic of Korea.

Dr Yoon points out that policies such as the one to reduce South Korea's greenhouse gas emissions by 30% over a 12-year period can only make a difference if rolled out and publicly accepted nationwide.

In the last few years, South Korea has also introduced numerous countrywide projects, such as the nation's Green R&D Financing Plan, in which 2% of its total GDP is invested into future environmental and waste solutions; and its Global Green Growth Institute think tank, which brings together expertise from both public- and private-sector bodies.

Dr Yoon argues it is because of such national initiatives that South Korea is now at the forefront of the environmental and waste R&D markets.

However, by rolling out plans on a national level, governments are always at risk of slowing down the decision-making process. This is the view expressed by the Honourable Mr Edgardo J. Angara, Senator, Republic of the Philippines, who states that because of the size of many Asian nations, plans to tackle environmental and waste challenges on a nationwide scale often get caught up in bureaucracy and indecision.

Mr Angara uses the example of the Philippines's water sector in support of his point. "We have 33 agencies involved in water management and supervision, and there is no single institution that takes responsibility. And, as a result of having 'too many cooks', almost 20% of the population still has no access to potable water, and around 24% of our people have no access to sanitation facilities," says Mr Angara.



(From left to right) His Excellency Dr Yoon Jong Soo, Vice Minister of Environment, Republic of Korea, His Excellency Dr Vivian Balakrishnan, Minister for the Environment and Water Resources, Republic of Singapore, the Honourable Mr Edgardo J. Angara, Senator, Republic of the Philippines, and Ms Pamela Cox, Vice President, East Asia and the Pacific, World Bank

CLEARLY, TO ACHIEVE GREATER RESOURCE EFFICIENCY, ASIA NEEDS ROBUST POLICY PLANNING, FASTER APPROVAL PROCESSES AND QUICKER PROJECT IMPLEMENTATION

Mr Angara believes the solution to these challenges is to put systems in place that allow for quicker government decision-making and that speed up implementation times.

Clearly, to achieve greater resource efficiency, Asia needs robust policy planning, faster approval processes and quicker project implementation.

However, what solutions actually exist in order to encourage greater resource efficiency?

SOLUTIONS

Many believe that the key to increasing resource efficiency is to change the behaviour of citizens and their perception of waste.

One such supporter of this theory is His Excellency Dr Vivian Balakrishnan, Singapore's Minister for the Environment and Water Resources, who believes that getting the public to value resources is key to enabling waste reuse.

Dr Balakrishnan claims that not subsidising resources leads to wiser usage of consumer goods — including food, clothing and power — and teaches citizens to greater value resources and waste.

"Resource consumption is not an ideological topic, it's a fact of life," asserts Dr Balakrishnan.

"Subsidies for energy, electricity and water – all these lead to wasteful consumption and ultimately place a strain on public infrastructure," the minister says.

"These actions uncover a certain ethic in your society, in that everything is scarce and that everything is precious, and you therefore don't waste it," continues Dr Balakrishnan.

However, educating citizens by means of national policies will not be enough to get them to think differently, argues Mr Angara.

South Korea has been successful in educating both the public and businesses on the benefits of responsible waste disposal



"You've got to roadshow every major policy throughout your country and explain what it is that you are trying to change extremely carefully, as well as harness the power of the media," says Mr Angara.

"I believe this approach will stick, because the constituencies themselves will have been consulted," says the senator.

Other methods of educating the public and businesses on responsible waste reuse include charging for waste collection, putting the responsibility of defunct goods collection on the very companies that manufactured them, and fixing a high price to valuable resources.

All three are currently proving successful in South Korea, as Dr Yoon explains.

"Extended producers' responsibility, which makes producers collect and recycle disused goods they have manufactured, is very successful in South Korea. As too is our volume-based trash system, which makes households pay the cost of the waste they discharge," claims Dr Yoon.

Dr Yoon also says that imposing the right price on resources is critical in support of waste reuse. However, Dr Yoon and many others believe that changing the perception of the public is only half of what needs to be done in enabling nations to be more resource efficient.

The other half is the development of new technologies that emit less pollution and allow for greater recycling.

Dr Yoon points out that in South Korea, designing products to be recycled following use is critical in achieving greater levels of resource reuse.

"Eco design at production has proved to be highly significant in our effort to raise resource circulation," claims Dr Yoon.

This has been successfully implemented in a range of consumer goods, including food packaging, white goods, electronic equipment and plastic items. According to Dr Yoon, other initiatives in South Korea that have proved to be environmentally enhancing include the use of renewable energy and electronic transportation systems. This is also the case across several Asian countries. According to Mr Angara, the Philippines not only has "the most impressive renewable energy portfolio in Southeast Asia", but has rolled out its own green transport system, known as the 'e-cycle' – a battery-driven tricycle. This aims to replace noisy and polluting motorcycles.

Likewise in China, the government is actively trying to reduce the need for transport by locating housing, recreational amenities and shops near places of work. China also encourages alternative fossil-fuelled transport, such as electronic vehicles, bicycles and electrical trams.

However, technology can also serve as a means of tracking waste generation, as proposed by the Government of Singapore. It suggests that by 2014, large waste generators – such as hotels and shopping malls – must report data on levels of disposed waste.

The purpose of this system is to publicly expose the amounts of solid waste produced by these organisations.

The path to resource efficiency is not an easy one for any Asian nation to undertake. Today, governments across Asia are faced with a multitude of choices relating to whether decisions are made locally or centrally, the level of bureaucracy required, and the involvement from businesses and the general public. However, what is clear is that there is no 'one-size-fits-all' solution, and that cities and countries must make rational decisions that are robust and thorough, and which consider all viable options available.

It is evident that the enablers of resource efficiency lie with people and technology. Behaviour change is key to prevent unnecessary and widespread waste, whereas technology can make waste reuse easier and more cost-effective. There is much work to be done by all nations and their stakeholders in order to make Asia more resource efficient.

ENVIRONMENTAL, HUMAN AND ECONOMIC EQUALITY

MS HELEN CLARK, UNITED NATIONS DEVELOPMENT PROGRAMME ADMINISTRATOR, ARGUES THAT DEVELOPMENT CANNOT BE SUSTAINABLE UNLESS IT INCORPORATES ALL ASPECTS

Human and economic development cannot progress without addressing environmental concerns. The reverse is also true. Ms Helen Clark, former Prime Minister of New Zealand and Administrator of the United Nations Development Programme, argues that the global human, economic and environmental ecosystem must be equitable in order to be sustainable.



Ms Helen Clark

The word 'development' should always be associated with sustainability. Ms Helen Clark, United Nations Development Programme (UNDP) Administrator, former Prime Minister of New Zealand and Chair of the United Nations Development Group, argues the case for public, private and civil organisations placing environmental considerations at the heart of their development strategies. "We can't plan for a sound economic and social future unless we take the environment very seriously indeed," claims Ms Clark.

Climatic or environmental changes have a disproportional impact on the many people living in highly unequal societies in extreme poverty. The stresses placed on the planet's economic, social and environmental ecosystems will only be exacerbated in the future. If these institutions are irreparably damaged, then further development will be impossible without a cohesive, integrated response that tackles both gross inequity and environmental sustainability.

As head of the UNDP, Ms Clark has to take both a short- and long-term view of sustainable development. She is an advocate of the Millennium Development Goals (MDG), which all 193 UN member states have signed up to achieving by 2015. The eight MDGs focus on three key areas: improving infrastructure, increasing social, economic and political rights, and raising the basic standards of living worldwide.

While Ms Clark views them as a little light on environmental concerns, she believes they have undoubtedly been a success: "They are few in number, are measurable, have a time and target, and are very easy to communicate."

She argues that the post-2015 development agenda will need to be defined more by the overlap of economic and environmental sustainability. World Bank figures show that the number of people living in extreme poverty has been halved, but future goals must deal with people living in poverty and with outright hunger. "There are still too many women dying in pregnancy or childbirth, we are still not reaching gender empowerment targets, and the sanitation goal is a bit of a sad story," says Ms Clark. There are successes though, and Ms Clark points to almost every child being in school, "But we have to get to the last mile."

"THE LEGACY OF POLLUTION AND GREENHOUSE GASES IS ONE BEQUEATHED TO US BY THE OLD INDUSTRIALISED AND DEVELOPED WORLD"

These problems cannot be tackled successfully if environmental, human and economic challenges are dealt with in silos. "The environmental pillar must be built up more, but we cannot ignore the core issue of a lack of opportunity and outright deprivation." In Ms Clark's view, "We have to go to zero on extreme poverty and hunger, if we are to have any hope of building a more just world."

"Equity and sustainability are linked," says Ms Clark, and economic development cannot be at the expense of human development. Both Ethiopia and the Cambodian capital, Phnon Penh have balanced economic growth with environmental protection and basic human needs.



According to Ms Helen Clark (left), governments and businesses need to take the environment more seriously. Seated right is Professor Tommy Koh, Ambassador-at-large, Ministry of Foreign Affairs, Republic of Singapore

FUTURE SOLUTIONS

Likewise, both developing and developed countries must do their share to 'green' their economies. In Ms Clark's view, there is a heavy responsibility on the latter to get this right: "The legacy of pollution and greenhouse gases is one bequeathed to us by the old industrialised and developed world." For developing countries, the means of implementation is important. As Ms Clark puts it, "Don't just tell us what to do, there's a need for support." This can come from developed countries carrying through their commitments to reduce emissions, or providing the finance required to build institutions within these countries.

The binding inter-government agreements between the world's member states to carry out these sometimes tough tasks can be difficult to both reach and sustain.

Few environmental protocols emerged from Rio+20, and Ms Clark admitted that the Commission on Sustainable Development (CSD), the body powered with implementing the various recommendations that came out of the 1992 Earth Summit in Rio, had struggled. It had not achieved its stated outcomes and failed to produce consensus or agreement.

The CSD was to be replaced by a high-level forum on sustainable development within the UN Economic and Social Council (ECOSOC) – the body that coordinates the economic, social and related work of various UN agencies. However, Ms Clark argued that genuine change is often now effected at a sub-national level. Local administrations, civil society and non-government organisations are integrating across the economic, social and environmental streams of sustainable development. Rio+20 saw around 700 voluntary agreements signed between these non-national government players, amounting to about US$500 billion.

This bottom-up approach will be crucial, but local and national governmental leadership must also show acumen to lead and the ability to cooperate. A green economy can only be achieved through an inclusive approach that lifts people out of poverty as it also plants the seeds for sustainable development.

Ms Helen Clark spoke with Professor Tommy Koh, Ambassador-at-Large, Ministry of Foreign Affairs, Republic of Singapore, at a joint forum held between the CleanEnviro Summit Singapore, World Cities Summit and Singapore International Water Week.

ABOUT HELEN CLARK

Ms Helen Clark was appointed the Administrator of the United Nations Development Programme (UNDP) in April 2009. She has overseen an array of humanitarian, ecological and innovative projects across the UNDP's four areas of work: poverty reduction, democratic governance, crisis prevention and recovery, and environmental and sustainable human development.

Prior to assuming this role, Ms Clark was the 37th Prime Minister of New Zealand and served with distinction for three consecutive terms, from 1999 to 2008. In 2006, *Forbes* ranked her the 20th most powerful woman in the world.

Ms Clark is also the Chair of the United Nations Development Group, a committee consisting of the heads of all UN funds, programmes and departments working on development issues.

DEALS
AT CLEANENVIRO SUMMIT SINGAPORE 2012

TOTAL NUMBERS

Total value of announcements for projects awarded, tenders and investment:

S$101.5 MILLION

An additional

S$500 MILLION

worth of potential deals under discussion.



From left to right: His Excellency Dr Vivian Balakrishnan, Minister for the Environment and Water Resources, Republic of Singapore, Mr Andrew Tan, Chief Executive Officer, National Environment Agency, Republic of Singapore, Ms Janet Ang, Managing Director, IBM Singapore, and Mr Jeff Rhoda, Vice President, Public Sector, Growth Markets, IBM Corporation

ANNOUNCEMENTS

OVER
S$10 million

The value of a MoU between Eco-Wiz and three Asian and European companies. Eco-Wiz is a Singaporean food waste management company that has taken its technology into the Asian and European waste markets. Eco-Wiz technology recycles food waste into reusable water.

S$13 million

A collaboration between the NEA and IBM to develop advance modelling and predictive capabilities to better manage key environmental concerns in Singapore, including air quality, extreme weather events, dengue outbreaks and food poisoning incidences.

S$15 million

Singapore's first District Pneumatic Refuse Conveyance System (DPRCS) at Marina Bay.

WTE deal between a Singapore company and an undisclosed client

S$61.5 MILLION

QUOTES

"Having had first-hand experience of balancing economic growth with environment sustainability, Singapore offers various good practices and lessons to be replicated in developing countries so as to overcome environmental challenges and promote resource efficiency."

Dr Arab Hoballah, Chief of Sustainable Consumption and Production, UNEP

"The events [CleanEnviro Summit Singapore and WasteMET Asia] gave us a good indication of the potential growth areas in the waste management sector. It also provided us with a springboard into Asia, allowing us to extend our business reach."

Mr Ho Nyok Yong, President, Samwoh Corporation

"[WasteMET Asia] was a beneficial platform that allowed us to meet with many high-level trade visitors and showcase our leading technology."

Mr Greg Brereton, Business Development Manager (Environment), Steinert Australia

BEYOND FACT AND FIGURES OVERCOMING TODAY'S WASTE CHALLENGES

Asia's waste challenges are plentiful, yet what can be done to both curb and better manage increasing levels of waste within the region? Mr David Newman, Mr Antonis Mavropoulos, Dr Paul Brunner and Dr Arab Hoballah offer solutions to today's foremost challenges.



WTE not only incinerates waste to mere particles, but also creates much-needed electricity

The facts look bleak. By 2025, solid waste in Asia is expected to reach 1.8 million tonnes per day. By this time, a further 500 million people across the region are expected to migrate to its rapidly expanding cities. These urban centres will contribute over half of the world's greenhouse gases by 2032.

As a result of this, many experts in this field believe that in order to successfully tackle Asia's environmental problems, we must first focus on cities.

Today, 52% of the global population – more than 3.6 billion people – has no access to waste management. In addition, urbanisation has been growing at a rate that is 30% faster than that of sanitation. This is notably the case within South Asia.

On top of this, global population is set to grow by a further 50% by 2050, with global GDP set to increase by 200%. Asia will play a major role in both growth rates, and in China alone a further 300 million people are predicted to migrate to the nation's burgeoning cities.

"What we need are new solutions for urban centres that go beyond traditional engineering and logistics," claims Mr Antonis Mavropoulos, Chair of the International Solid Waste Association (ISWA).

Mr Mavropoulos points out that one such solution is the development of underground waste management systems, as trialled at Singapore's Changi Airport where waste is transported from its source to treatment facilities by means of an underground pneumatic refuse conveyance system.

However, today's waste challenges will require not only new technological innovations, they will also require a significant shift in human behaviour.

"WE HAVE NOT CONCENTRATED ENOUGH ON CHANGING CONSUMPTION BEHAVIOURS, AND GOVERNMENTS AND PRIVATE ENTERPRISES MUST DO MORE TO ACHIEVE THIS"

Dr Arab Hoballah, Sustainable Consumption and Production Branch Chief, Division of Technology, Industry and Economics, United Nations Environment Programme

"The first thing we must do is change the way we as humans view and treat waste," claims Mr Mavropoulos.

Dr Arab Hoballah, Chief of the Sustainable Consumption and Production Branch in the Division of Technology, Industry and Economics, at the United Nations Environment Programme, agrees with this, adding that both the public sector and private companies must contribute to this change.

"We have not concentrated enough on changing consumption behaviours, and governments and private enterprises must do more to achieve this," claims Dr Hoballah. However, how can this be achieved?

Dr Paul Brunner, Head of the Institute for Water Quality, Resources and Waste Management at Vienna's University of Technology, believes the catalyst for behavioural change lies with increasing our knowledge of waste and urban metabolism.

"The goals of waste management can only be reached if we have a better understanding of total metabolism in our cities," claims Dr Brunner.

He argues that the problem with waste is not about a lack of space for landfills, but rather that the materials we extract from the Earth's crust are substantially different once they have been consumed by "anthropogenic, man-made metabolism." Dr Brunner states the problems of today's waste simply lie in our lack of understanding of substances.

Dr Hoballah too believes that only once we have mastered our understanding of waste, will the world be equipped to enable behavioural change.

"At the end of the day, we don't know enough about global waste and the data we have are mostly estimates. We need to improve our knowledge if we are to take really effective action and change the way society treats waste," claims Dr Hoballah.

Having the knowledge is one thing. Making change happen is another issue, as pointed out by Dr Hoballah.

"Each and every one of us talks a lot about changing behaviour and consumption patterns, but we don't change a lot. This is because it is very difficult to do and it requires much courage," says Dr Hoballah.

"Getting society to be more responsible will prove challenging, but it is essential," he claims.

Dr Hoballah believes that whilst governments are responsible for key decisions, policymaking and regulation, they must work in tandem with market forces, including private enterprises and non-government organisations (NGOs) in order to enable behavioural change.

Conversely, some experts believe the solutions to today's challenges lie more in how waste is actually managed, rather than with consumer behaviour. Mr David Newman, ISWA Vice President, shares this view.

"The world produces four billion tonnes of waste per year. This figure is not the problem – the problem is how we manage it," claims Mr Newman.

Mr Newman calls for greater participation in recycling and better management of landfill. With regards to the latter, Mr Newman points out that 800 million tonnes of greenhouse



Aluminum cans can be reused an unlimited number of times, and the process of recycling them takes a mere 60 days

gases are produced every year because of current waste management techniques, with 93% of these emitted from landfill. Therefore one solution to this is to capture the methane released from landfills and convert it into energy.

According to Mr Newman, recycling can play a major role in curbing emissions. He adds that one tonne of paper saves 2.3 cubic metres of landfill space and enough energy to power a single American household for six months.

In addition, Mr Newman puts forward greater usage of waste-to-energy (WTE) technology, which not only incinerates waste to mere particles fit for landfill, but also creates much-needed energy.

Mr Newman also suggests greater use of composting than is currently exercised around the world. This will additionally reduce the amount of waste that goes to landfills and will allow for the recovery of precious nutrients needed for fertiliser.

However, Mr Newman is quick to acknowledge that, above all, the waste sector needs greater investment, and notes the lack of capital currently being injected into the sector, particularly in emerging markets.

"One of the key factors that is holding us back is how little overseas development aid is going into waste management projects," he says.

Cities now house over 50% of the world's population and are where the majority of solid waste challenges occur



Mr Newman claims that only around US$400 million a year out of US$160 billion of overseas development aid goes into waste, and notes that the World Bank acknowledges waste to be as serious a problem as climate change, thus reaffirming his call for greater investment.

The benefits of better waste management to society are plentiful, as explained by Mr Newman.

"There are so many benefits to come from waste management, whether economic, environmental or social. It provides more jobs, guaranteed urban hygiene, less emissions, and new energy sources that replace fossil fuels," he says.

However, Mr Newman points out that effective waste management cannot be achieved in the short term and must be viewed by policymakers, the private sector and other stakeholders in the longer term.

It is clear from the above viewpoints that there are four calls to action in order to best tackle Asia's growing waste challenges.

Firstly, governments, NGOs and the private sector must focus on cities, as for the first time in history they are home to the majority of the world's population and are where the overwhelming majority of waste challenges reside.

Secondly, more investment in waste management studies is required, as many experts in this field believe nations still do not know enough about urban metabolism.

Thirdly, public- and private-sector organisations must educate the public on responsible waste disposal, otherwise volumes of waste will increase perhaps beyond the overly high amounts that are currently forecast over the next 40 years.

Lastly, the world needs to better manage its waste, whether through recycling, WTE or composting.

What is also clear is that it is the responsibility of everyone to tackle the world's waste challenges. This not only includes government agencies, businesses and international think tanks, but also the general public that currently live their lives ignorant of the many challenges associated with responsible waste management.

FACTS & FIGURES

By 2050,
50%
of the world's population will live in the Asia Pacific region



Global urbanisation is growing
33%
faster than sanitation infrastructure



More than half of the world's slum dwellers live in Asia



THE NEED FOR STRONG LEADERSHIP

MR ANDREW TAN, CHIEF EXECUTIVE OFFICER, NATIONAL ENVIRONMENT AGENCY, REPUBLIC OF SINGAPORE SPEAKS

Looking back to 1965 and the birth of Singapore, the nation encountered many challenges that are no different to those experienced by multinational companies (MNCs) operating in today's complex and dynamic economic environment.

At the time, the nation had high unemployment levels, an increasing population, and no basic industries to rely on. The leader of this new nation was then Prime Minister Mr Lee Kwan Yew, who was charged with the task of building Singapore. Instead of following the path of other Asian cities, Mr Lee felt it was essential for the nation to differentiate itself. Yet, how could this be achieved when Singapore looked like every other city in the region?

Singapore in the late 1960s was dirty, heavily polluted and underdeveloped. Therefore, one of the first things Mr Lee did was to clean up Singapore, make its appearance greener, and invest in infrastructure. This allowed potential investors to get to their meetings on time or ahead of schedule, as well as enjoy pleasant scenery.

As Mr Lee travelled far and wide promoting Singapore around the world, his pitch was that the country had a system that was well-organised, efficient and worked – unlike many of its competitors in Asia at that time.

Mr Lee saw the importance of environmental stewardship, not only as a means to differentiate Singapore from other nations, but also as a strategic and competitive advantage – this was long before the term 'sustainability' crept into present-day vocabulary.

Today, many CEOs see sustainability as a key differentiator, thus sharing Mr Lee's views from over 40 years ago.

STRONG LEADERSHIP

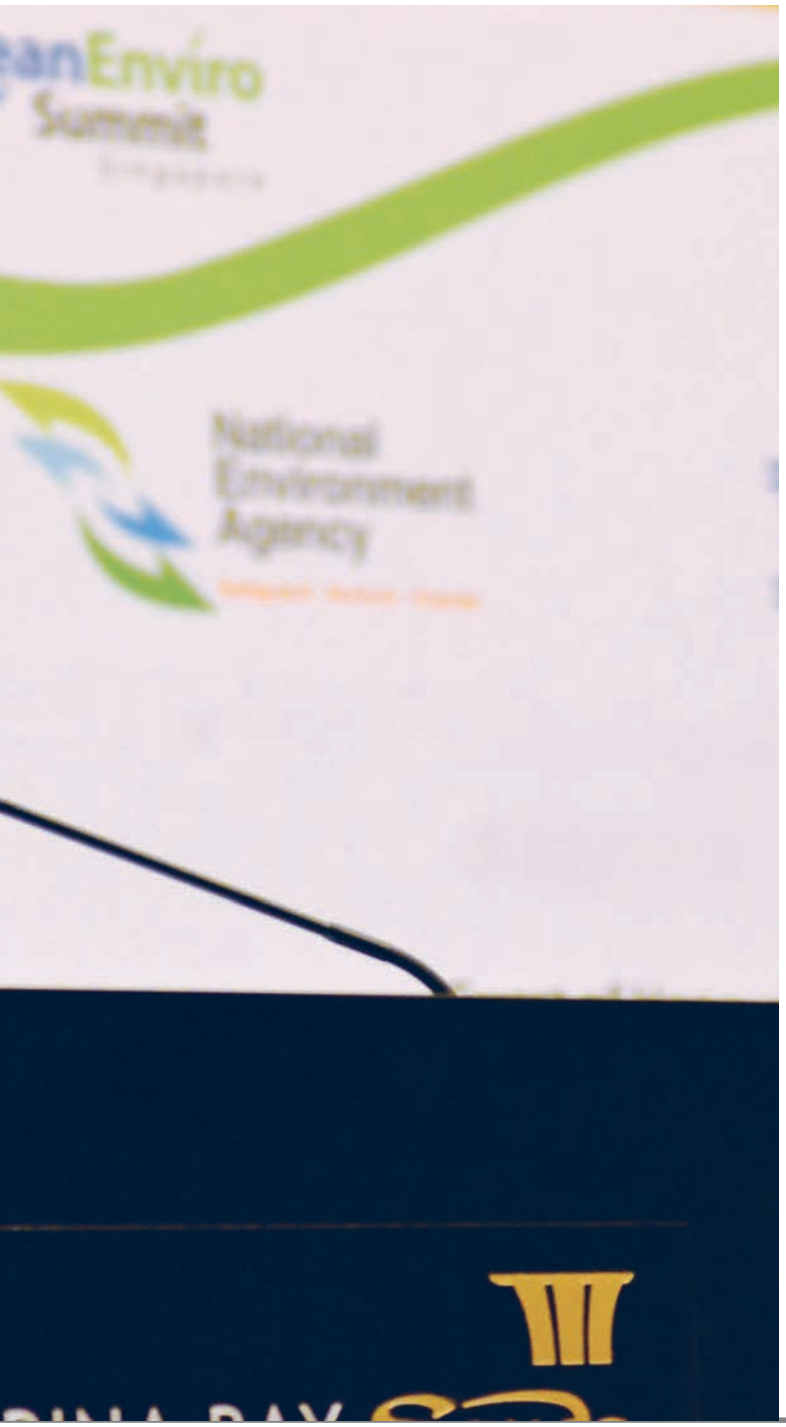
Mr Lee enabled change because he believed in it. Similarly, any sustainability agenda today needs strong leaders who believe in its cause, because without such people organisations will never be able to implement real change.

Even such leaders will not convince everyone, especially within large organisations. However, we now operate in a world where the public is well informed and can access information on the actions of organisations. Therefore it is important to have a transparent, open and accountable system that makes information readily available to both internal and external stakeholders.

Indeed, this is an opportunity, rather than a hindrance. For example, regulators such as Singapore's National Environmental Agency



ANY SUSTAINABILITY AGENDA TODAY NEEDS STRONG LEADERS WHO BELIEVE IN ITS CAUSE, BECAUSE WITHOUT SUCH PEOPLE ORGANISATIONS WILL NEVER BE ABLE TO IMPLEMENT REAL CHANGE



are leveraging social media and the public's access to information. This can provide the agency with critical information on what the public thinks about its actions and policies.

For some time, customers have had the opportunity to air their views on products they deem not to be environmentally friendly. However, there are now other considerations, as companies must now pay greater attention to the way their products are decided, made and distributed.

Organisations can easily put the best systems and incentives in place, but in order for these to work effectively they must have people who believe in them.

OPPORTUNITIES

As leaders in both business and government passionately push forward today's sustainability agenda, opportunities for all sectors will arise. New and sustainable products will drive technological advancements that are of greater benefit to society and corporations economically, environmentally and socially.

However, governments, private companies and industry organisations must work together to enable such innovation and it is only through collaboration against the backdrop of strong leadership that all parties will achieve this feat.

It is because of strong leadership that today's foremost companies and countries are where they are today.

FACTS & FIGURES

In some Asian cities, expenditure on municipal waste reaches

40%

of the municipality's total operating budget



China is the world's largest generator of municipal solid waste, producing almost

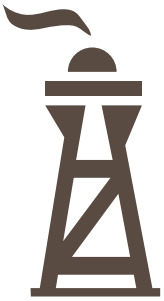
150

million tonnes

of waste every year, with the amount growing at an annual rate of eight to 10%

18%

of all anthropogenic methane emissions are due to landfill



THE BEST SOLUTION
RECYCLING, LANDFILL OR WASTE-TO-ENERGY?

Foremost solutions to today's solid waste challenges include landfills, recycling and waste-to-energy. Which one is best suited to tackle the complex and somewhat overbearing problems associated with solid waste? Is there one standout solution, or is each better suited to a particular physical environment?



Today's recycling market includes paper and cardboard, glass, metals, plastics, organic waste and e-waste

Recycling, waste-to-energy (WTE) and landfills have three common attributes. Firstly, all three are highly diverse and incorporate an array of different methodologies and technologies. Secondly, all three solutions have been around for some time and have become highly sophisticated over the last few decades. Finally, all three have a multitude of advantages and disadvantages.

RECYCLING

Today's recycling technologies allow a plethora of materials to be reused. This includes paper and cardboard, glass, metals (such as aluminium cans and containers), polyethylene terephthalate (PET) plastics (such as those used for packaging plastics and bottled water), organic waste, and e-waste (including white goods and electronic equipment).

There are many benefits to recycling. First, there is a buoyant market for most forms of recycling, which in turn creates jobs and high amounts of revenue and income. Second, recycling helps conserve the world's limited resources and is energy efficient. Third, recycling is deemed and publicly understood to be environmentally friendly.

However, despite these benefits, some experts argue against recycling due to a number of factors. These include claims that certain products – such as e-waste and vehicle parts – leak toxic materials into the environment if not treated correctly. Indeed,



The latest WTE technologies are able to generate significant amounts of electricity

this is something the US Environmental Protection Agency warns against.

The cost of collecting and processing of recyclable materials is generally higher than those associated with other solutions, such as landfills. For example, a report published by the National Institute of Environmental Health Sciences highlighted in 2011 that in the US, a tonne of waste costs around US$28 to landfill, whereas it costs US$147 to recycle the same amount.

Recycling also generates pollutants and emissions through the burning of fossil fuels by vehicles used to collect and transport the recyclable materials. In addition, recycling uses a great deal of water during processing. Plus, some materials, also referred to as residual waste, are simply not recyclable, such as ferrous metals and certain types of plastics.

In order for recycling to be a viable solution to today's waste challenges, three conditions must exist. First, there must be systems and infrastructure in place prior to the commencement of any recycling programme. Second, citizens must comply and fully engage with such a programme. Third, there must be a market for recycled materials.

WTE IS WIDELY ACKNOWLEDGED TO BE ONE OF THE MORE SUSTAINABLE FORMS OF ENERGY

In many of today's emerging markets, only one or two of these prerequisites are met, which makes recycling a difficult solution to fully implement.

WASTE-TO-ENERGY

For markets where recycling appears to not be the best solution, WTE is a viable alternative, as this technology uses waste that would otherwise go to landfills, and can make use of both recyclable and non-recyclable materials. Most notably, this solution is particularly beneficial to countries with limited land space, such as Singapore or smaller European countries.

In addition, the cost of waste as fuel is significantly cheaper than other forms of energy, such as coal and natural gas.

As a result, WTE is widely acknowledged to be one of the more sustainable forms of energy, simply because people will always have waste. In fact, old landfills can now be mined and materials at these sites turned into energy.

However, WTE plants are expensive to construct, as they need high-end capital equipment that captures unwanted emissions let off during the combustion of waste. In fact, WTE is over twice as expensive to construct and operate when compared to some landfill technologies.

In addition, once the waste is incinerated, ash is produced that also needs to be disposed of. Critics of WTE claim that one day this ash will fill landfills and so WTE is no different to this technology. However, in recent years, the ash has been used in concrete production, cement manufacturing, soil amendment and as a means of adding extra grip for vehicles on snow-laden roads in cooler climates.

Some experts also argue that incineration discourages people from recycling, though many others say this is not the case.

LANDFILL

The concept of landfill has been around for millennia, and until the middle of last century they were no more than holes in the ground where solid waste was disposed of in a disorderly, unhygienic and often dangerous manner. Although such traits still exist in developing economies, there are four principal forms of the modern-day landfill.

The first is known as the controlled landfill. This includes simple non-engineered techniques such as making the dump secure with perimeter fencing and a single point of entry and exit. Waste is levelled and covered daily in order to control odours and the presence of vermin.

THE BENEFITS OF LANDFILL TO MUNICIPALITIES ARE MULTIPLE AND IT IS CONSIDERED BY MANY EXPERTS TO BE THE MOST ROBUST OF SOLID WASTE SOLUTIONS



The second form is the engineered landfill, which adopts the same principles as above, but with the addition of a low-permeability cap placed over its top in order to prevent infiltration of rainwater that, without which, can create leachate. The cap also captures methane that is released from waste as it decomposes, and this is burnt via a gas flare. Although at first this may be seen as being environmentally damaging, in truth the flare converts methane into carbon dioxide, with the former being 20 times more damaging to the environment than the latter.

The third type of landfill is the sanitary landfill, which is lined with low-permeability minerals or man-made synthetic materials such as high-density polyethylene. It too has in-situ infrastructure that treats methane and leachate. Yet, it differs from an engineered landfill in that the methane emitted from its waste is used for renewable energy in the form of electricity generation and heat.

The fourth type of landfill is the bioreactor landfill, which incorporates all of the above, yet consists solely of organic material. It capitalises on the process of fermentation that occurs when biological material decays. This is a relatively new technology that is still being researched, particularly in North America. However, a number of vendors have made this technology commercial, such as France's Veolia Environmental Services.

The benefits of landfill to municipalities are multiple and it is considered by many experts to be the most robust of solid waste solutions. Emissions are controlled and greenhouse gases put to good use. The fact that they have security gates means that all waste entering a landfill site can be monitored, making operation and regulation relatively easy.

Landfills can be sited on derelict land areas, such as disused quarries, and can accommodate changes in waste composition with no detrimental impact on the process, unlike other methodologies such as recycling or WTE.



The sharing of information in the form of case studies and expert opinions has become a key part of the CleanEnviro Summit Singapore



The Semakau Landfill is arguably one of the waste world's greatest engineering achievements

Perhaps of most significance, the upfront investment required for a landfill is lower than that needed for other technologies. However, landfills are not without faults. In developing countries they are largely uncontrolled and are environmentally damaging. In addition, those that capture methane only confine around 50% of what is actually emitted.

From a long-term perspective, the biggest challenge associated with landfills is physical space. On average, landfills use an area two kilometres in length by one kilometre-wide every 10 years. Hence, for smaller countries or those with competing interests for land, landfills are considered an inappropriate solution with which to tackle increasing levels of solid waste.

THE BEST SOLUTION?

In short, there is no best solution, and all three methodologies are suited to a particular situation and set of circumstances.

For example, for more developed nations with limited land space, WTE appears to be the best solution. For countries where recycling infrastructure, public buy-in and commercial markets for recycled materials exist, recycling is highly viable. For developing nations where competition for public money is rife, engineered and sanitary landfills appear to be best suited.

However, selecting a solution is but a small part of today's solid waste challenges. Defining clear objectives and setting goals that are environmentally and economically sound, as well as being sustainable in the long- term, is equally challenging for governments and the private sector alike.

Over time, the world will run short of metal ores, minerals and a host of non-renewable natural resources. It is in this light that policymakers will do well to invest in recycling and reuse innovations.

This article is based on a debate staged at WasteMET Asia – an industry event part of the CleanEnviro Summit Singapore 2012 – between Mr Hermann Koller, Managing Director of the International Solid Waste Association, Mr Derek Greedy, President of the Chartered Institution of Wastes Management, and Dr Nicholas Themelis, Professor of Chemical Metallurgy at Columbia University, New York City.

FACTS & FIGURES

98%
of CEOs in the Asia Pacific view sustainability as essential to their company's strategy

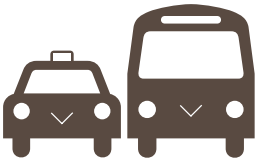


Most Asian towns and cities use open dumps and only about

10%
of solid waste ends up in properly engineered and managed landfills

In China,

26%
of all greenhouse gas emissions come from transport



23%
of the population or 850 million people in the Asia Pacific region practice open defaecation

THE BUSINESS CASE FOR SUSTAINABILITY

FINDING 'TRUE NORTH' FOR COMPETITIVE ADVANTAGE

How do CEOs view sustainability? What are their current challenges, where do commercial opportunities lie, and what more can be done by today's leaders to truly meet the sustainability agenda? Mr Peter Lacy, Asia Pacific Managing Director of Accenture's Sustainability Services, discusses a new report published by Accenture.


Mr Peter Lacy

Three years ago I led the world's largest study of CEOs on sustainability, commissioned by Dr Ban Ki-moon, the United Nations Secretary General.

The study surveyed some 766 CEOs from all around the world, spanning 25 industry sectors and more than 100 different countries. It also comprised 50 face-to-face interviews with leading Global Compact CEOs and a further 50 face-to-face with other senior stakeholders.

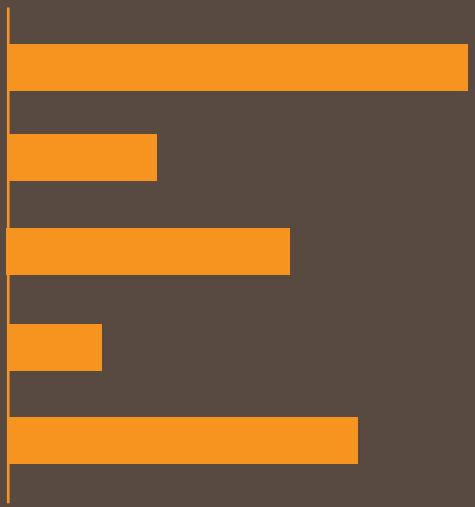
The most striking finding was that 93% of CEOs told us that they saw sustainability to be of paramount importance for the future success of their business. Interestingly, that rose to 98% in the Asia Pacific region – a fact that flies in the face of those who thought sustainability was a quasi-North American-European issue.

Across the entire Asia Pacific, we saw real concerns, not just for future climate change and carbon emissions, but also for present-day challenges such as air pollution, contaminated water, increasing levels of waste and energy shortages.

For Western CEOs, many of these issues are at arm's length. Yet, in Asia they are right on the doorstep of their leaders.

THE BUSINESS CASE FOR SUSTAINABILITY

The below are some findings published in Accenture's 2012 sustainability report, entitled *Long-Term Growth, Short-Term Differentiation and Profits from Sustainable Products and Services,* which examines the responses of 766 CEOs globally.



98% of CEOs in the Asia Pacific region see sustainability to be of paramount importance to the future success of their business

37% of CEOs globally say they cannot keep up with demand for sustainable products

56% of CEOs globally expect greater demand for sustainable products in the future

26% of CEOs globally use social media and digital technology to engage with customers regarding sustainable products

60% of CEOs globally believe they can charge a premium for sustainable products



CEOs across Asia view sustainability as key in winning new customers

THE GAP BETWEEN MAKING A WELL-MEANT EFFORT AND MAKING REAL CHANGE IS SIGNIFICANT

Those CEOs surveyed were leaders of companies that had already committed to sustainability and were members of the United Nations Global Compact. But, they were not from companies that had committed their entire business models to sustainability, such as The Body Shop or Ben & Jerry's.

They were the likes of Unilever, HSBC, Alcoa and Tata – big businesses that are currently making it into the mainstream.

The reasons why sustainability mattered so much to these companies were because of four drivers: new growth opportunities; cost reduction; risk management; and building all-important reputation, both with stakeholders that allow them to operate, as well as those that enable them to innovate and grow.

What was also telling was that 10 years ago the biggest issues in sustainability were public awareness and being able to craft a strategy.

Today, sustainability is about execution – it is about moving from strategy to implementation, and from policies to action.

CEOs are now not wrestling with strategic direction, they are struggling to find plausible ways to drive sustainability into their core business.

And, in ways that act as accelerators of profitability and growth, rather than a break on these same things.

This includes new thinking on how they embed sustainability into their supply chains, drive cost efficiencies, and promote environmental stewardship and social welfare.

A recent study conducted by GlaxoSmithKline, Oxford University's Smith School and Accenture looked at the implications of climate change on human health, as well as the outcomes for the pharmaceutical sector in the medium to long-term. They found that global shifts in the world's weather are changing the patterns of diseases, such as the return of malaria to Europe, where it has been absent for centuries.

Thus, this is just one of the long-term issues – for example, human health that will impact the daily activities of mainstream businesses. The implications for pharmaceutical R&D and the future demand for medicines are one way that sustainability will reshape the future of an industry that at first glance isn't immediately the central sector from a sustainability perspective.

However, only a select few are 100% committed to the execution of a robust sustainability agenda. And, if we look at the remaining 60,000+ multinational corporations globally, there is still a very, very long way to go.

The gap between making a well-meant effort and making real change is significant.

Looking back over the last 20 years, we have seen much progress in sustainability that has included initiatives at the government, industry and individual company level, which were simply not evident in the early 1990s.

Thankfully, sustainability today is no longer 'window dressing', peripheral corporate social responsibility, philanthropy and a 'nice to have' that sits on the edge of a business's operations – it is becoming mainstream.

But, this is not enough.

The signs tell us that at every level we are falling behind (what the science and economics tell us we need to do) to achieve a healthy and prosperous global economy for the future.



Sustainability is becoming more mainstream, but experts argue that this is not enough and more must be done by companies to meet today's needs

THE WORLD NEEDS TO DECOUPLE FOSSIL FUELS FROM INDUSTRIAL PRODUCTION, ECONOMIC GROWTH AND PROSPERITY

I like to compare the world's current efforts and what is actually needed to the difference between true north and magnetic north. When navigating short distances, using either magnetic or true north will let you get to the same place.

Yet, when travelling several thousand miles, the difference between using magnetic north and true north could result in being hundreds of miles off course – regarding sustainability, the world is currently not heading to true north and will be quite some way off where it needs to be by 2050.

Recently, the International Energy Agency released a report showing that in 2011 carbon emissions hit a record all-time high, and were up 5% on 2008 levels.

Likewise, we have seen some fantastic work carried out by the Forest Stewardship Council (FSC) that looks to bring forests under a more tightly regulated and sustainable framework. The organisation has grown 40% year-on-year since 1993. But, currently less than 3.4% of the world's forests are under the FSC mark and deforestation is growing at a faster pace per hectare than the reach of the FSC.

We need to be very clear about what the real challenges and business opportunities are. And, in order to address these we must face an industrial revolution.
The world currently produces over 40 gigatonnes of greenhouse gases a year. Yet, in order to a avoid a 2°C increase in the Earth's temperature – what many scientists deem to be the start of a catastrophic shift in climate change – we must reduce our emissions to around 20 gigatonnes per year by 2050.

But, with the global economy set to treble in 40 years, greenhouse gases are set to rise to over 60 gigatonnes per year, which would likely prove apocalyptic for our planet.

In order to counter this, we need to decrease our fossil fuel usage by around 14% every year, for the next 40 years – the world needs to decouple fossil fuels from industrial production, economic growth and prosperity.

The world faces a new industrial revolution. And this begins with sustainable companies that are led by visionary and pioneering leaders, who not only view sustainability as a moral obligation, but who see this agenda as an opportunity for economic, social and environmental prosperity.

Those who see this as a once in a generation business opportunity – perhaps even a once in an era opportunity to create a step-shift for the better in the way we live, work and play; and in turn who see this as an opportunity to create a new wave of economic opportunity for the companies that provide the products and services required for the transition; those who innovate new ways of thinking about production and consumption – will not only demonstrate the power of market forces to deliver outcomes for citizens and consumers, but will create sustained competitive advantage in a new era of sustainability. That's the prize of setting your compass to 'True North' innovation on sustainable business.

The content of this article was transcribed from the forum entitled 'Profitability Through Sustainability' staged at the CEO Forum on Sustainability – a co-located event at the CleanEnviro Summit Singapore 2012.



The declaration document that emerged from Rio+20 called for worldwide action on solid waste

BEYOND RIO+20

ADVANCING THE 3RS AND RESOURCE RECOVERY

From a waste management perspective, industry experts and government leaders alike deemed the Rio+20 meeting highly successful. What facets of waste management were addressed during the event, and what must governments and private companies do to ensure the recommendations outlined in the event's declaration are enacted?

Rio+20 witnessed a large number of positive outcomes for the waste and environmental sectors. The event's outcome document, *The Future We Want*, which aims to be a guide for governments and companies' future policymaking, made numerous commitments in support of sustainable waste management practices.

Of most significance, the document deemed waste to be an "emergency" on an international scale. It argued the robust management of waste to be a key prerequisite for achieving global sustainability.

Of the 283 articles that made up the declaration, nine were specifically dedicated to the promotion of sound waste management on a global scale.

THE RIO+20 DECLARATION

In brief, the declaration urged organisations to support the management of waste by means of applying the 3 Rs:

- **Reduce**
- **Reuse**
- **Recycle**

The Rio+20 declaration placed a particular emphasis on the use of waste as a resource, whether in the form of energy or recycled materials. The declaration also called for the appropriate management of wastes that pose notable challenges – including e-waste and other residual waste forms – and encouraged less waste surrounding post-harvest produce within the food supply chain.

It also asked countries to take all possible measures to prevent unsafe treatment of hazardous wastes and called for long-term funding in the waste and environmental sector – particularly in emerging economies.

UNEP'S DR MATTHEW GUBB BELIEVES INDUSTRIAL SYMBIOSIS CAN BE A CATALYST FOR REAL CHANGE



ENABLING ACTION
The above calls were positively received, particularly by industry bodies that have campaigned hard for decades to bring waste management to the forefront of the international agenda. Yet, what must be done to ensure these recommendations are enacted accordingly?

Mr CRC Mohanty, Environmental Programme Coordinator at the United Nations Centre for Regional Development, believes there to be four key actions that must be undertaken in order to achieve goals set forward in the Rio+20 declaration.

1. Zero waste economy
In this concept, it is forbidden to dispose of solid waste in landfills. Some European countries – such as Belgium and Austria – have already implemented this, which has led to significantly less consumption of natural resources. This is not just the case with water, energy and metals, levels of emissions and air pollution also dropped notably, while human health and economic activity improved.

2. Setting limits
Limiting certain resources and consumer products per capita, such as one car per family, could enhance the 'zero waste economy' concept. As could the possible introduction of a closed-loop economy, where waste goods are exchanged for new or other products. The belief behind such radical thinking is that these actions will encourage waste reuse and an overall reduction in human consumption.

3. Education
Levels of waste in high-income countries are four times that of lower-income nations. Mr Mohanty points towards education as a means of changing the behaviours found in wealthier nations.

4. Industrial symbiosis
Effective environmental solutions require greater cooperation between governments, private companies and industry groups. Mr Mohanty believes 'industrial symbiosis' – the sharing of services among diverse industrial actors – is the perfect solution to bring all three parties together.

This is a view that is also shared by Dr Matthew Gubb, Coordinator of the Secretariat for the Strategic Approach to International Chemicals Management at the United Nations Environmental Programme.

Dr Gubb believes industrial symbiosis can be a catalyst for real change. He believes that it can only be enacted through incremental steps that incorporate rigorous planning, test piloting, and large-scale policies that are ultimately designed for the long-term. Dr Gubb argues that there is no 'one-size-fits-all' solution to bringing all parties together. He calls on the various stakeholder groups to find the solution that best suits their needs.

THE ROLE OF TECHNOLOGY
Mr Mohammad Helmy, Vice Chairman of the Indonesian Solid Waste Association, claims technology has an equally important role in ensuring the realisation of the Rio+20 declaration.

Mr Helmy argues for the introduction of cost effective, small-scale emission and pollution reduction technologies, which require low upfront investment, operational and maintenance costs.

Likewise, Mr Helmy claims small-scale and cost effective technology solutions for recycling, and energy and



Mr Miezah Kodwo from the Ghana Institute of Waste Management shares Africa's many challenges and opportunities



The UN recently declared solid waste to be an emergency on an international scale

water reduction to be key in achieving the waste objectives set out in the Rio+20 declaration.

FINANCING

As pointed out by Mr Jeff Cooper, President of the International Solid Waste Association, the goals of Rio+20 must be supported by financial and economic mechanisms. Dr Gubb agrees, saying there must be a coherent and comprehensive plan laid out and agreed to by international governments. This will allow financial multilaterals – such as the World Bank or Asian Development Bank – to provide adequate funding.

Mr Cooper also points towards the expansion of Nationally Appropriate Mitigation Actions, or NAMAs – policies and actions that countries undertake as a means of financing anti-greenhouse gas emission programmes — as an alternative source of funds for the waste sector. This would bring financial and technological assistance to significantly more waste-mitigating schemes around the world that would otherwise not take place.

THE WAY FORWARD

A large part of Rio+20 focused on the 'green economy'. Despite widespread criticism in the international media for alleged 'inaction' on the part of both governments and the UN, waste is now a priority for all parties. Waste is now not only a precious resource, it is an important economic realm that provides jobs and income for both governments and private corporations.

The highest-profile outcome to emerge from Rio+20 was the establishment of the Sustainable Development Goals. These incorporate all three pillars of sustainable development: economic prosperity, social awareness and environmental stewardship. It is now clear that waste as a resource can deliver on all three.

What governments, the private sector and industry leaders alike agree is that in order for the Rio+20 to be a true success, the text of the declaration must move from black and white into real action. This can happen by means of conventional and unconventional practices. Some of these may be easy, while others are expected to be more challenging.

As the above industry insights show, such action only seems likely when all stakeholders have fully bought into sustainable waste management. Support in the form of adequate financing and cutting-edge technology will also be critical to the success of tomorrow's waste agenda.

This article is based on a presentation entitled 'Advancing the 3Rs and Resource Efficiency in the context of Rio+20' staged at WasteMET Asia – an industry event part of the CleanEnviro Summit Singapore 2012 – and featuring Mr CRC Mohanty, Dr Matthew Gubb, Mr Mohammad Helmy and Mr Jeff Cooper.

ABOUT RIO+20

Rio+20, officially called the United Nations Conference on Sustainable Development or UNCSD, was the third international conference on sustainable development aimed at reconciling the economic and environmental goals of the global community.

The summit was hosted by the city of Rio de Janeiro, Brazil, in June 2012 and marked the 20th anniversary of the 1992 United Nations Conference on Environment and Development (UNCED) and the 10th anniversary of the 2002 World Summit on Sustainable Development (WSSD) staged in Johannesburg, South Africa.

The 10-day summit, which culminated in a three-day high-level UN conference, was organised by the United Nations Department of Economic and Social Affairs and included participation from 192 UN member states. This included 57 heads of state and 31 heads of government, as well as private-sector CEOs, and leaders of non-government organisations from all corners of the world.

THE BIG INTERVIEW

WITH MR TAY LIM HENG, CHIEF EXECUTIVE OFFICER, KEPPEL INTEGRATED ENGINEERING


Mr Tay Lim Heng

Waste incineration has been around for some time. How has it changed and why is it now highly competitive and favoured by many governments?

Mr Tay Lim Heng: Some countries have found it convenient and inexpensive to send waste to landfills directly. However, with growing populations, rapid urbanisation, rising transport costs, and competing demands for land use, dedicating large areas to landfills is becoming untenable and unsustainable. Furthermore, if not properly treated, waste buried in landfills release unwanted substances, including greenhouse gases and toxins, into the environment.

For countries that have space constraints, waste incineration achieves a very high rate of landfill diversion. This can be done in a compact facility. For example, the Keppel Seghers Tuas Waste-to-Energy (WTE) Plant is built on a land space of only 1.6 hectares – one of the most compact WTE facilities in the world. It has the capacity to treat 800 tonnes of solid waste a day to generate 22 megawatts (MW) of electricity, and can achieve a landfill diversion rate of up to 90%.

With rising fuel prices, WTE plants also provide a source of green energy. The Domestic Solid Waste Management Centre (DSWMC) in Qatar, operated by Keppel Integrated Engineering (KIE) can produce up to 50MW of power, of which about 30MW can be exported to the national grid – this is enough energy to power a medium-sized residential community.

What are the latest technologies, and how do they work?
Some of today's more cutting-edge technologies include the Dano Drum, flue gas cleaning systems and multi-stage grate technology.

The Dano Drum is a horizontally mounted steel cylinder that shreds, mixes and sorts refuse into different streams for either recycling, biological processing, combustion with energy recovery or landfill disposal.

For example, at the DSWMC, organic waste is removed and taken to an anaerobic digestion and composting plant where it is composted to produce soil enhancers for use in agriculture and landscaping. Recyclables such as plastics (near infrared separator), and ferrous (magnetic separator) and non-ferrous (eddy current separator) metal are separated and sent for recycling. Residual waste that cannot be recycled or composted – known as Refuse Derived Fuel (RDF) – is sent to a WTE plant to generate electricity.

The by-products of this combustion process, in the form of ash, are treated in-situ at the DSWMC and disposed in engineered landfills. By this stage, the waste volume is reduced by more than 95%.

Today's WTE plants are equipped with state-of-the-art flue gas cleaning systems that meet the most stringent of emission regulations.

Keppel Seghers recently retrofitted the Senoko WTE plant in Singapore with our flue gas treatment technology, which is significantly more effective at removing harmful and acidic gases, dust and particulates.

Yet, at the core of all Keppel Seghers' WTE plants is multi-stage grate technology. Here, combustion of waste takes place on either air-cooled or water-cooled grates that are engineered to withstand high temperature and abrasion. In addition, through sliding and tumbling movements, the grate tiles control the feeding rate whilst ensuring thorough mixing and burning of the waste.

Keppel Seghers' grate treatment system is the only such technology in the world that incorporates separate and independently controlled actuators for waste-mass transport and optimal waste mixing.

Where do you see the most promising markets for waste incineration and WTE?
With many countries facing dramatic population growth, rapid urbanisation and resource scarcity, modern WTE is establishing itself as an attractive solution to sustainable development. At KIE we see specific opportunities in China, the Middle East, Europe, and Brazil.

China is expected to invest up to 180 billion yuan in the urban waste disposal sector. The growing demand for effective waste management solutions will continue to drive the implementation of WTE technologies across the country,



which will be supported with favourable policies.

In China, KIE is exploring new opportunities beyond the provision of the combustion and flue gas treatment systems for WTE plants to further strengthen the company's track record. This could be in the form of partnerships with local governments and companies, which allow all parties to synergise their knowledge and expertise.

In the Middle East, Qatar stands out for its efforts to deliver a high standard of living for its people through sustainable development.

With the DSWMC and Doha North Sewage Treatment Works, the Qatari government is spearheading the transformation and development of public infrastructure throughout the country. The two projects by Keppel Seghers stand as showpieces of the Qatari leadership's foresight and strong commitment to setting international standards of environment protection.

KIE sees opportunities for further collaboration with the Qatari government and other Gulf Cooperation Council, or GCC, countries in the building of similar world-class facilities that serve the region.

The European nations are recognised as being at the forefront of the WTE movement. Legislative drivers by the EU have propelled considerable growth in the region's WTE industry. As a result of the EU's Landfill Directive, which was officially implemented in 2001, many EU member states are under pressure to divert residual waste from landfill to meet their 2020 landfill diversion obligations. The shortage of capacity has resulted in the planning and commissioning of an increasing number of WTE facilities over the past decade, of which KIE sees much growth potential.

We also see Brazil moving away from its reliance on landfills. Brazil's National Solid Waste Policy was published in 2010 and aimed to advance the waste management industry by means of reuse and recycling facilities, and the generation of energy from waste.

The Dano Drum is a horizontally mounted steel cylinder that shreds, mixes and sorts solid waste



What do you believe must be done to lower levels of solid waste throughout Asia?
Many governments have implemented bills and public awareness programmes to encourage the reduction of waste, the reuse of items and the recycling of materials to shape people's behaviour on waste disposal.

While these efforts need to continue, the changing of attitudes and behaviours will take time. Furthermore, increasing waste generation will continue to place pressures on our existing waste management infrastructure. This is especially so in Asia, where many cities are growing rapidly.



Accessible waste solutions will drive the implementation of innovative technology across the Asia Pacific

The private sector too can play a role in augmenting waste management through the provision of technology.

Do you think there will come a time when mankind can do away with landfill and rely solely on future incineration and recycling technologies?
We have learned that in order for any solution to be widely adopted, it must not only be technologically sound, but also commercially viable. Whilst we will see newer technologies that may offer better recovery rates, the technologies that are unable to gain market acceptance are unlikely to be successful.

What are the benefits of forums such as CleanEnviro Summit Singapore and WasteMET Asia to solution providers such as KIE?
Singapore does not enjoy the luxury of having space for landfill, but this has driven us to develop a very efficient and affordable waste management system. We are pleased to share our experiences with other countries that may face the same issues as the nation once did.

With public-private partnerships gaining ground, KIE is keen to offer its experience and technical expertise to help solve the waste problems of Singapore and beyond.

The CleanEnviro Summit Singapore and WasteMET Asia have allowed us to conduct candid exchanges at the global network level, and the company looks forward to seeing the connections made at the conference evolve into strong partnerships.

BEHIND CLOSED DOORS

ASIA'S PREMIERE ENVIRONMENTAL REGULATORS FORUM

THE CLEAN ENVIRONMENT REGULATORS ROUNDTABLE
The inaugural Clean Environment Regulators Roundtable (CERR) was staged at the CleanEnviro Summit Singapore and attended by delegates from 11 countries, the World Bank (co-organiser), United Nations Centre for Regional Development (UNCRD), and United Nations Environment Programme (UNEP). Countries in attendance included Australia, Belgium, Brunei, Cambodia, China (including Hong Kong), Indonesia, Malaysia, Singapore, Thailand, United Arab Emirates, and Vietnam.

CONCEPT
Environmental regulators worldwide face similar regulatory and enforcement challenges when dealing with environmental pollution and waste management challenges. When presented with unfamiliar industrial projects or complex industrial environmental problems, regulators often have no one to turn to for advice or consultation.

Until the debut of CERR, there was no such forum where the regulators could help each other by sharing their experiences and technical information. As a result, regulators often have difficulty finding solutions to the challenges they must overcome, despite such solutions potentially having been developed and successfully implemented in other countries.

The inaugural CERR provided just this – a unique platform for government regulators to share problems and solutions with like-minded regulators from other nations.

Only regulators and like-minded partners were invited, as this closed-door forum encouraged frank

ENVIRONMENTAL REGULATORS WORLDWIDE FACE SIMILAR REGULATORY AND ENFORCEMENT ISSUES WHEN DEALING WITH ENVIRONMENTAL POLLUTION AND WASTE MANAGEMENT CHALLENGES

Attendees of the inaugural CERR spanned 11 different countries, the UN and the World Bank





sharing of both unsolved problems and solutions to other regulators in a secure and confidential atmosphere.

Regulators were thus able to learn from each other's experiences and use their collective wisdom to develop better solutions to numerous present-day challenges.

OUTCOME

Participating countries shared case studies covering an array of topics. These included successes and challenges in household waste management in Belgium; enforcement and management challenges of a landfill in Australia; implications of regulations on vehicle emissions in Thailand; meeting water quality challenges in Indonesia; and the challenges associated with controlling pollution in the UAE – a country currently experiencing rapid economic and population growth.

As a result of such dialogue, participants made arrangements for further collaboration after the forum.

Singapore expressed interest in Belgium's advanced source monitoring programme that can provide quicker measurements of pollutants, such as dioxins in the stacks of refuse incineration plants.

In addition, Singapore and Hong Kong currently face similar challenges in controlling construction noise in highly built up areas, and subsequently agreed to collaborate further to find better solutions to this challenge.

Singapore also shared its schemes to minimise the number of vehicles with smoky emissions on the road, enforce laws against illegal dumping of waste, manage construction noise, control flare emissions from petrochemical facilities, as well as manage solid waste at Semakau Landfill. The city-state also took the opportunity to encourage other countries to consider using ground flares to solve the problems of excessive light, noise and air pollution from petrochemical plants.

THE WAY FORWARD

Upon conclusion of the roundtable, participants unanimously agreed the session to be highly beneficial in developing solutions to today's waste and environmental problems.

They also expressed a great deal of interest in attending future sessions and requested that the meetings be held annually instead of biannually.



Mr Kamran Khan, Program Director of the World Bank's Global Infrastructure Finance Center of Excellence, voices his views on how best to overcome today's environmental issues

FACTS & FIGURES

90%
of South Asia has no access to solid waste management facilities



More than 800 thermal WTE plants currently operate in around

40
countries globally

In China, by 2025 a further

300 million
people are expected to move from rural to urban areas



Asian cities are likely to contribute more than half the rise in greenhouse gas emissions by

2032

FROM THE FLOOR

PICTURES, VIEWS, AND REVIEWS FROM THE 2012 CLEANENVIRO SUMMIT SINGAPORE AND WASTEMET ASIA EXHIBITION AND CONFERENCE



"Sanitation is a really important issue in Nigeria and other parts of Africa. I hope that the CleanEnviro Summit will allow people to share their expertise on this and other issues."

Mr Dalhatu Altaji Shehu,
Commission for Environment,
Sokoto State Government, Federal
Republic of Nigeria



"My work involves policies and regulations. The lessons learnt from other countries' regulatory experiences will be important in helping to reduce pollution."

Ms Inar Ichasana Ishak,
Ministry of the Environment,
Republic of Indonesia



"The most important thing about the CleanEnviro Summit is that delegates exchange their experiences and talk about their visions for the future."

Mr Che-Ping Sherman Fong,
Environmental Protection Department,
Hong Kong Special Administrative
Region of the Poeple's Republic of China



Ms Grace Fu (seated centre), Second Minister for the Environment and Water Resources and Second Minister for Foreign Affairs, Republic of Singapore, learning to handle a tractor on a simulator as an amused audience looks on at the WasteMET Asia Expo.



Mr Kenichi Sasaki, President of Dowa Eco-System Co. Ltd, engaged in an interview conducted by the Japanese media at the CleanEnviro Summit Singapore.



CleanEnviro Summit delegates take a ride in a Volvo simulator, which veers almost 90° left and right to demonstrate the safety of the Volvo vehicle over uneven terrain.



Mr Stan Krpan, CEO of Sustainability Victoria, sharing Australia's many experiences at the Clean Environment Leaders Plenary.

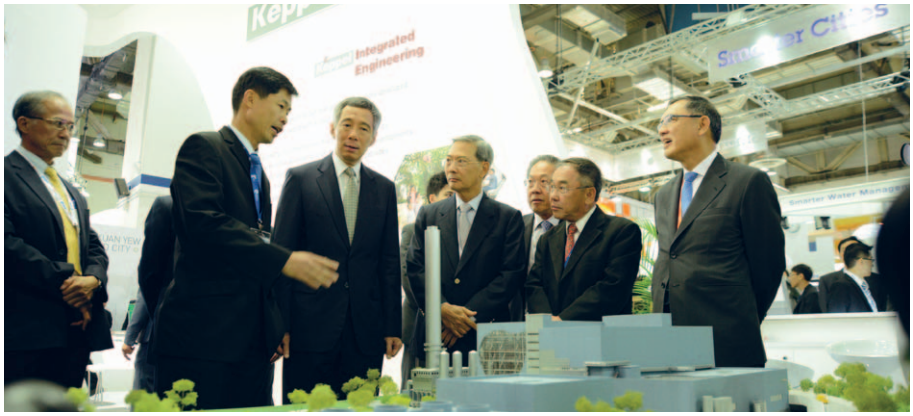


Mr Andrea Blanquet, R&D Engineer of Malaysia-based STREAM Environment Sdn Bhd, demonstrates the mechanics of the Waste Shuttle rubbish collection system, which uses 20% less power than a conventional system.



Mr Rudi van der Weerd, export manager of manufacturer Royal Dutch Bammens, explains how OTTO Waste Systems' Underground Waste System is a more practical mode of garbage disposal by taking up less above-ground space with its subsurface storage container.

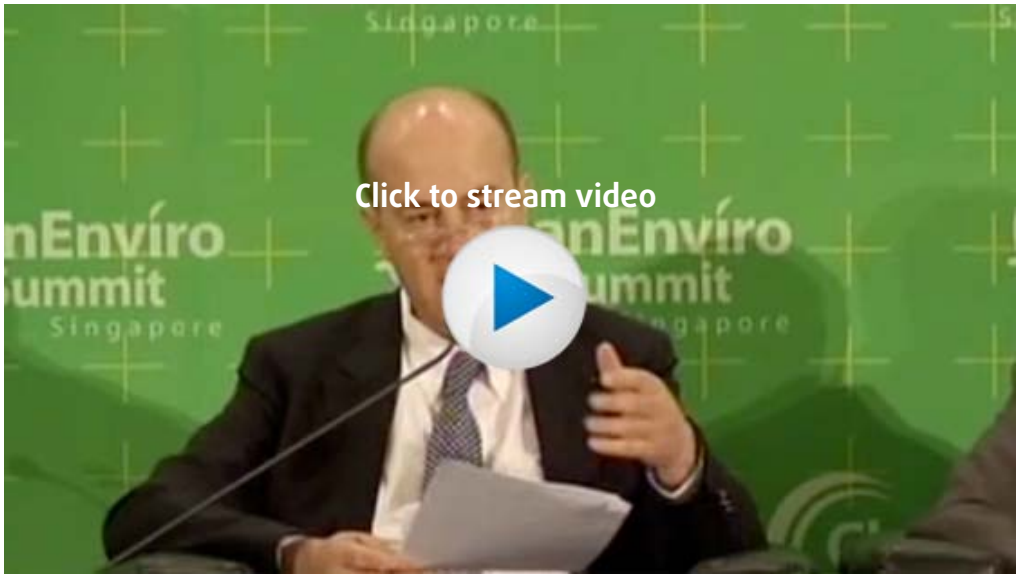
Mr Tay Lim Heng (second from left), Chief Executive Officer of Keppel Integrated Engineering, introducing its landmark Domestic Solid Waste Management Centre in Qatar to the Prime Minister of Singapore, Mr Lee Hsien Loong (centre), as Dr Lee Boon Yang (fourth from left), Chairman of Keppel Corporation, looks on.



EXPERT INSIGHTS

VIDEO COVERAGE FROM THE CLEANENVIRO SUMMIT SINGAPORE

THE LATEST CUTTING-EDGE WASTE TECHNOLOGIES



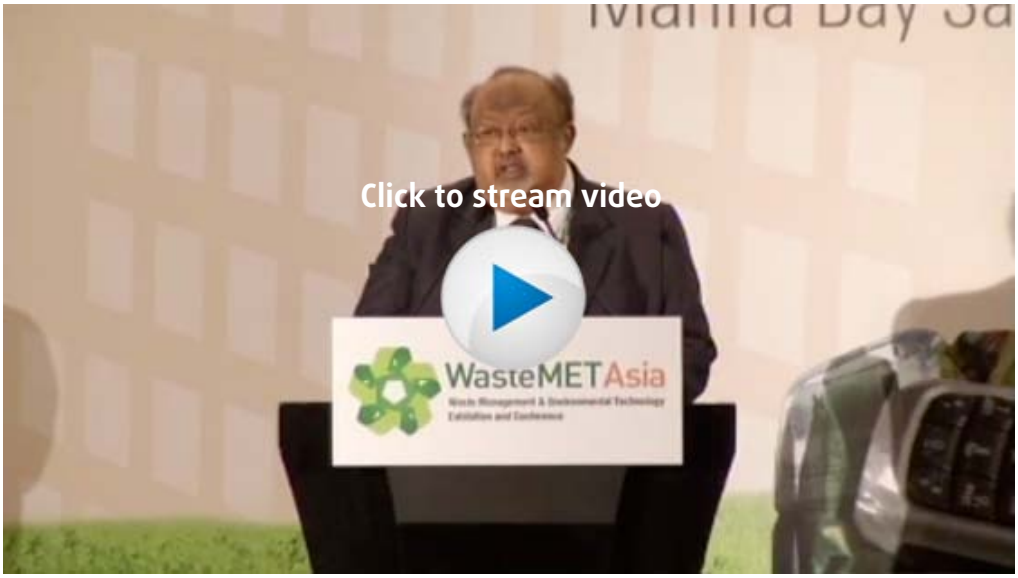
Mr Jérôme Le Conte, Chief Executive Officer of Veolia Environmental Services. discusses new technologies in the fields of household waste collection, waste sorting, the recovery of exotic materials, and energy recovery.

WHAT GOVERNMENTS MUST DO TO ATTRACT PRIVATE-SECTOR INVESTMENT IN WASTE



Dr Arab Hoballah, Chief of the Sustainable Consumption and Production Branch, Division of Technology, United Nations Environment Programme, shares his views about what governments must do to attract greater private-sector investment in waste with Mr Chris Tobias, Assistant Director of the Singapore Environment Institute.

SOLUTIONS TO INDIA'S MUNICIPAL SOLID WASTE CHALLENGES



Mr Prasad Mosak, Executive President of the Environmental Management Centre in Mumbai, offers solutions to present-day solid waste challenges in India, including what must be done to treat plastics, construction refuse, electronic waste and household hazardous waste.

HOW GOVERNMENTS MUST ADDRESS GREENHOUSE GASES AND CLIMATE CHANGE



His Excellency Dr Vivian Balakrishnan, Minister for the Environment and Water Resources, Republic of Singapore, discusses the challenges associated with greenhouse gas emissions and climate change.

THE SINGAPORE DIFFERENCE

TOWARDS ZERO WASTE

Very few countries have adopted the 3Rs of waste management as wholeheartedly as Singapore, particularly in the Asia Pacific region. On top of this, the concepts of reduce, reuse and recycle are supported by a robust regulatory framework, high amounts of investment – from both the Singaporean government and private companies – and a large number of public-private partnerships (PPPs) that merge industry expertise with civil knowledge.

This has not always been the case for Singapore, whose humble beginnings go back to the mid-1960s, when it was a developing nation characterised by pollution and underdeveloped infrastructure, and housed a population of around two million.

How times have changed. Much has been written about Singapore's meteoric rise to developed nation status in the space of a few decades, particularly with reference to the city-state's first Prime Minister, Mr Lee Kwan Yew, who saw a direct correlation between a clean environment and economic prosperity.

The process of cleaning up Singapore was not easy and to this day remains a challenge. However, its government has maintained a somewhat aggressive approach to solid waste management and relentlessly continues in its quest for zero waste.

Historians point out that Singapore's pursuit of the 3Rs was out of necessity rather than any other reason, such



Mr Lim Swee Say, former Minister for the Environment and Water Resources, Republic of Singapore, trials Veolia Environmental Services' GRIN (which stands for Grow your Recycling Incentives Now) initiative, which has achieved an 80% public participation rate



Photo: Courtesy of WDA Singapore

as political gain. The country now houses a population of over five million, who reside on a land area of approximately 710 square kilometres. Simply put, land is not in abundance.

In 2011, the city-state generated more than 17,800 tonnes of solid waste per day, which amassed to 6.5 million tonnes for the entire year. Successfully managing high levels of waste against the backdrop of limited land space has made Singapore unique in the international waste market. Indeed, it is yet another example of 'the Singapore difference'.

REDUCE

Waste minimisation is a core value of Singapore. Rather than the 'carrot and stick' approach that is currently used by many other nations across Asia, the city-state has chosen a more collaborative route to reducing levels of solid waste.

One such initiative is the Singapore Packaging Agreement (SPA), a government-led initiative launched in 2007 that aims to reduce product packaging, without compromising on health and safety issues. As of March 2012, the agreement spanned 135 organisations, with many of its signatories being global multinationals.

The results of the SPA have been impressive – over a four-year period spanning July 2007 to June 2011, the agreement saved the local economy just under S$15 million that would otherwise have been spent on both the production of packaging and the costs associated with its disposal.

Singapore continues to actively engage all industries to find ways to reduce waste generation, and even provides funding to companies that wish to rework processes in order to reduce levels of solid waste. The country's National Environment Agency (NEA) encourages corporations to take ownership of their own waste through afterlife management of their products and packaging. This applies to manufacturers, distributors, retailers, consumers, waste collectors and recycling companies.

REUSE

A large part of Singapore's success story has been the ability to turn waste into energy. Through its four operational waste-to-energy (WTE) plants, Singapore has become more energy independent, and the latest – the Keppel Seghers Tuas plant built in 2009 – is a fine example of how private corporations can work in tandem with the government, as it is operated by local company Keppel Seghers (part of the Keppel Corporation).

WTE technology is dominated by incineration, which turns waste into energy by means of steam turbines that are driven by the heat released during combustion. In 2010, Singapore's four WTE plants generated 1.17 million megawatt hours of electricity, contributing to around 2% of the nation's electricity consumption.



Photo: Courtesy of Keppel

Cutting-edge WTE technology has enabled Singapore to reduce its waste volume by over 90%

DUE TO TECHNOLOGICAL BREAKTHROUGHS, SINGAPORE IS ABLE TO RECYCLE MATERIALS THAT WERE ONCE DEEMED AS RESIDUAL WASTE

Other WTE technologies under consideration include gasification (converting gases emitted by organic waste into burnable fuel), pyrolysis (the thermal decomposing of organic wastes to produce transport or power generation fuels), and fermentation (where organic waste is fermented in order to produce fuels such as ethanol and hydrogen).

Deemed to be cost-effective and highly reliable, Singapore's incineration plants shrink waste volumes by around 90%. Discharge in the form of ash is then transported to the Semakau Landfill on the island of Pulau Semakau.

Arguably one of Singapore's greatest engineering achievements, the offshore Semakau facility is the first of its kind in the world, receiving 1,500 tonnes of incinerated ash and 500 tonnes of non-incinerated ash daily. With a capacity of 63 million cubic metres, the landfill will not only serve Singapore's waste-disposal needs until the year 2040, but will also create new land as a result of the landfill process.

Moreover, the island has become a major tourist attraction as it has maintained its rich biodiversity in perfect harmony with the landfill.

The upfront investment needed to build the Semakau Landfill stood at around S$610 million, and the project is yet another example of how committed the Government of Singapore is, to implementing robust long-term waste management solutions.

RECYCLING

Recycling has also received much attention. Singapore prides itself on its recycling rate of 58%. This is a significant increase from its rate of 40% as recorded at the turn of the century,



Ms Grace Fu, Second Minister for the Environment and Water Resources and Second Minister of Foreign Affairs, Republic of Singapore,and Mr Akio Yoshinari, Management Director of DOWA ECO-SYSTEM SINGAPORE (a subsidiary of DOWA group which provides a comprehensive range of environment protection services including industrial waste management and precious metal recycling)

and which remains far higher than other ASEAN nations. Due to technological breakthroughs, Singapore is able to recycle materials that were once deemed residual waste (waste that cannot be recycled). This includes slag, construction and demolition waste, and ferrous metals, where over 90% is today recycled.

Not content with its recycling rate of 58%, Singapore is aiming to increase this figure to 65% by 2020 – which it remains on course to achieving – and to no less than 70% by 2030. Meanwhile, other neighbouring Asian nations are struggling to meet recycling rates of around 30%.

In its bid to achieve these ambitious targets, Singapore has rolled out several initiatives. These include the formation of the Waste Management and Recycling Association of Singapore, the S$20 million 'Innovation for Environmental Sustainability Fund' that supports cutting-edge waste management R&D, and extensive educational drives aimed at educating both businesses and the general public — such as Singapore's Annual Recycling Week and National Recycling Programme.

Singapore has also facilitated numerous private recycling initiatives. One such venture is Veolia Environmental Services' GRIN (Grow your Recycling Incentives Now) programme, which rewards residents with numerous items based on the amount of recycled waste disposed of in specially designed recycling bins.

The initial results of this particular initiative were impressive. In the space of just six months during the scheme's piloting in 2010, monthly volumes of collected recyclable waste increased from around 100 tonnes to around 350 tonnes in select residential areas. The programme managed to achieve a participation rate of over 80%.

Due to its success, the scheme is currently being extended to offices in some districts in the city-state.

Another example of the nation's commitment to recycling is the precious and non-ferrous metals recovery plant operated by Dowa Eco-System Singapore. The plant, which was opened in tandem with the 2012 CleanEnvrio Summit, aims to recycle metals such as gold, silver and palladium by up to 50 kilogrammes each per year, according to Dowa Eco-System's Mr Jun Yamamoto.

SINGAPORE AS A GLOBAL HUB

The relentless pursuit of the 3Rs, supported by continued investment in R&D and the staging of cutting-edge fora such as the CleanEnviro Summit Singapore and WasteMET Asia, has ensured Singapore's status as a global hub for the waste sector.

As a result, many of today's leading vendors run their Asia Pacific operations out of Singapore and the city-state continues to attract leading academics, practitioners and industry think tanks to its shores.

Few countries in Asia – if not the world – are as active as Singapore in addressing today's foremost waste challenges and the subsequent creation of robust solutions to overcome these. At the heart of these activities is the NEA, which works in tandem with communities, other public sector agencies and private corporations, and has spearheaded numerous initiatives since its launch in 2002. The NEA's responsibility to ensure Singaporeans enjoy clean air, land and water is partially met by its commitment to reduce, recycle and reuse.

For decades, Singapore has shown the rest of the world that by valuing waste countries are able to become more sustainable and rely less on the world's depleting natural resources. Singapore's four WTE plants, the Semakau Landfill, the Singapore Packaging Agreement and the city-state's array of recycling initiatives are testament to this.

Due to successes and advances in this area, Singapore is now a global leader and pioneer in the pursuit of waste as a resource.

CLEANENVIRO SUMMIT SINGAPORE 2014

A MUST-ATTEND EVENT



The 2014 CleanEnviro Summit Singapore will be a must-attend event for industry and government leaders from all corners of the world



Due to the outstanding success of this year's CleanEnviro Summit Singapore, the National Environment Agency (NEA) is already planning the event's next showing on 1–4 June 2014.

Not only will attendees get to hear from industry and government leaders from all corners of the world, delegates will be able to network with other top professionals, as well as learn from the best practitioners who have tried and tested cutting-edge policies and technologies for an array of waste and environmental challenges.

BIGGER AND BETTER

The 2014 summit will not only follow up on the great successes from this year's event – including the Clean Environment Leaders Summit (see pages 1–3), the Regulators Roundtable to be co-organised with the World Bank (see pages 24–25), and the WasteMET Asia Exhibition and Conference (see pages 26–27) – it will also address other key environmental challenges, including air pollution, public health and public cleaning.

CleanEnviro Summit Singapore 2014 will be co-located with Singapore International Water Week and the World Cities Summit, highlighting the present-day importance of the synergy between water, energy and the environment. Thus, it will be a must-attend event for environmental and waste professionals from all around the world.

WHAT'S NEW FOR 2014...

More than half of the world's most polluted cities are in Asia and, according to the World Health Organization (WHO), air pollution contributes to around half a million deaths every year. If left unattended, by 2050 carbon dioxide emissions could reach three times those recorded in 2008. For many, this will indeed prove fatal.

The CleanEnviro Summit Singapore 2014 will provide guidance to governments, corporations and industry organisations on how best to prevent air pollution. This will include legislative, enforcement and monitoring measures currently used by governmental and private-sector market leaders.

PUBLIC CLEANLINESS

As Asia's cities grow at an unprecedented rate, the need to keep the region's streets, public areas and public amenities litter-free and clean has moved up on the agenda of urban policymakers.

The CleanEnviro Summit Singapore 2014 will address such topics as litter prevention, pest and stray animal control, and keeping public facilities clean and accessible. The event will provide case studies from stalwarts in this area from Singapore, Europe and North America.

CleanEnviro Summit Singapore

www.cleanenvirosummit.sg

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